

Trainings ▾

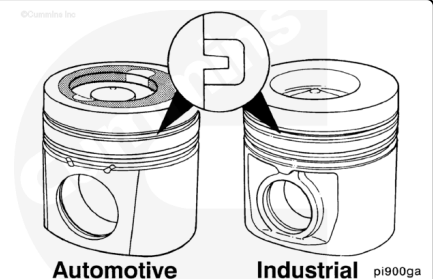
General Information

Front Gear Train

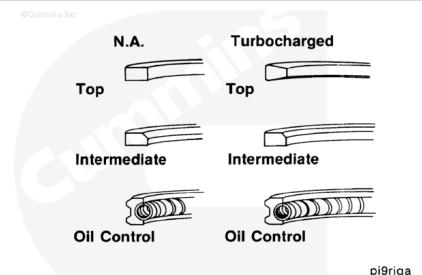
Piston features include high-swirl combustion bowl cast-aluminum body and three-ring grooves. The piston for turbocharged, turbocharged/aftercooled engines includes a Ni-resist insert with a keystone profile for the top piston ring.

Always check the part number to be sure the correct configuration is used during piston replacement.

In addition to the Ni-resist insert, automotive turbocharged engines also feature pistons with a hard-anodized combustion surface.

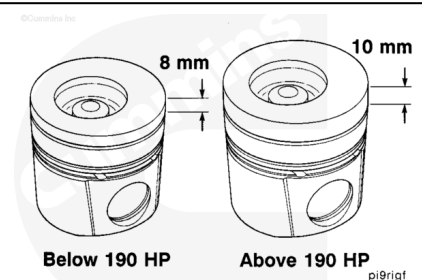


The piston ring sets are also different. While both sets consist of three rings, the top ring of the turbocharged/aftercooled set has a keystone profile that operates in a Ni-resist insert cast into the piston. The naturally aspirated top ring is square cut and operates in a groove machined into the aluminum piston.



The 1994 automotive pistons utilize different top ring locations. The 160-hp to 175-hp ratings use 8 mm crown to ring land pistons and 190 hp to 210 hp use 10 mm. The 1994 industrial pistons continue with the 14 mm ring position.

Note : Not all industrial two-valve pistons have a 6 mm top ring and a single Ni-resist insert, beginning in 2003.

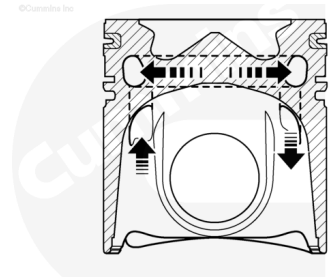


Rear Gear Train

The piston features:

- A centered combustion bowl for more efficient combustion
- An offset piston pin for reduced combustion noise

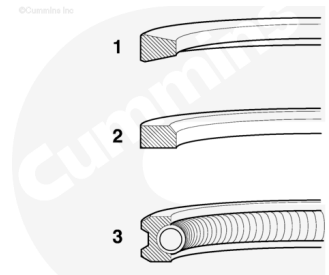
- An internal oil gallery for piston cooling. Oil is sprayed at the underside of the piston by directed J-jet piston cooling nozzles. When the piston is near BDC, oil is sprayed through the internal oil gallery for maximum piston cooling.



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The piston rings set consists of three rings:

1. The Top Piston Ring — which has a keyway profile that operates in a Ni-resist insert cast into the piston
2. The Intermediate Ring (second ring)
3. The Oil Control Ring.



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Preparatory Steps

- Remove the piston and connecting rod assembly. Refer to Procedure 001-054, Piston and Connecting Rod Assembly (</qs3/pubsys2/xml/en/procedures/40/40-001-054-tr.html>).



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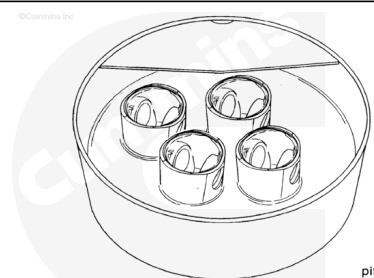
Clean and Inspect for Reuse

⚠ CAUTION ⚠

Do not use the bead-blast method to clean the pistons. The pistons will be damaged by blast material embedded in the aluminum.

⚠ CAUTION ⚠

Do not clean the pistons in an acid tank. Damage to the piston can occur.



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Soak the pistons in cold parts cleaner.

Note : Soaking the pistons overnight will usually loosen the carbon deposits.

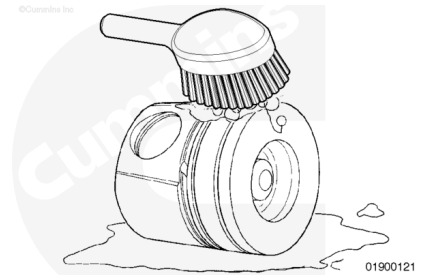
⚠ CAUTION ⚠

Do not clean the pistons in an acid tank. Damage to the pistons can occur.

⚠ CAUTION ⚠

Do not use a metal brush. A metal brush will damage the piston ring grooves.

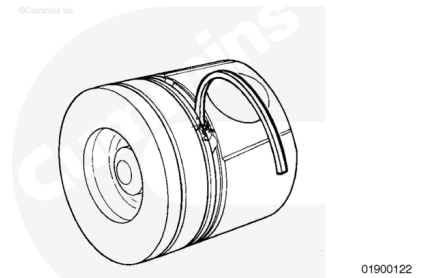
Wash the pistons in a strong solution of laundry detergent and hot water.



⚠ CAUTION ⚠

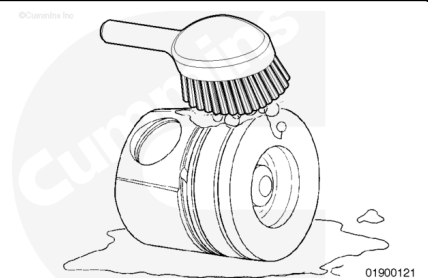
Do not use a ring groove cleaner and make sure not to scratch the ring sealing surface in the piston groove.

Clean the remaining deposits from the ring grooves with the square end of a broken ring.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ CAUTION ⚠

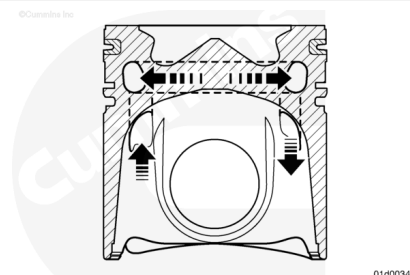
Do not clean the pistons and connecting rods in an acid tank. Damage to the pistons and connecting rods can occur.

Wash the pistons again in a detergent solution or solvent.

Rinse the pistons in clean, hot water.

Dry the pistons with compressed air.

Note : B4.5 RGT engines have an oil passage cast into the top of the piston for cooling purposes. When cleaning the pistons, make sure the oil passage is clean and free of debris.



Failure Analysis Inspection

Rear Gear Train

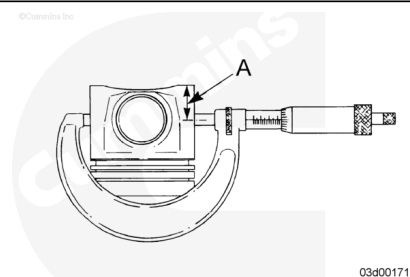
Inspect the piston for damage and wear to the skirt, pin bore, top, and ring lands.

Inspect the piston pin for damage and wear.



Note : To ensure accuracy, the following measurements should be completed with the components at room temperature, 20°C [68°F].

Measure the piston skirt diameter 21.4 mm [0.84 in] (A) from the bottom of the piston.



B4.5 RGT Standard Piston Skirt Diameter

mm		in
106.878	MIN	4.2078
106.892	MAX	4.2083

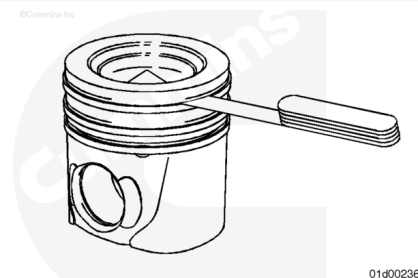
B4.5 RGT Oversize Piston Skirt Diameter

	mm		in
Oversize 0.5 mm	107.378	MIN	4.2275

	107.395	MAX	4.2280
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Measure the piston ring clearance. Use a new piston ring to measure the clearance in the ring groove.

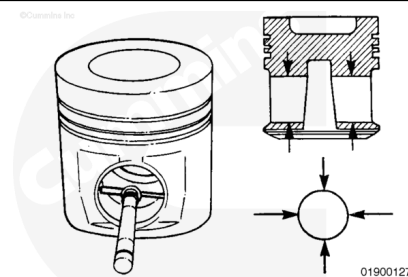
B4.5 RGT Piston Ring Clearance			
	mm		in
Intermediate	0.040	MIN	0.0016
	0.110	MAX	0.0043
Oil Control	0.040	MIN	0.0016
	0.085	MAX	0.0033



Note : The top piston ring clearance is not measured due to the type of piston ring used. The clearance can not be measured accurately with a typical feeler gauge.

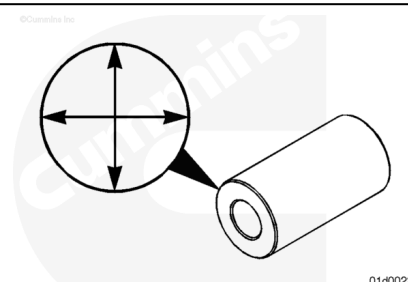
Measure the piston pin bore.

B4.5 RGT Piston Pin Bore		
mm		in
40.006	MIN	1.5750
40.012	MAX	1.5753



Measure the piston pin diameter.

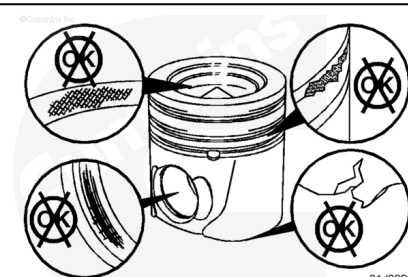
B4.5 RGT Piston Pin Diameter		
mm		in
39.997	MIN	1.5747
40.003	MAX	1.5749



Front Gear Train

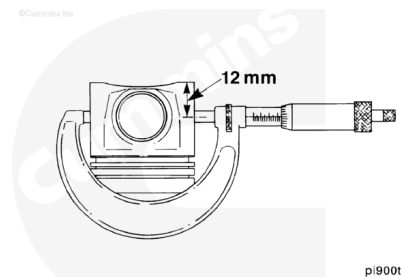
Inspect the piston for damage and wear to the skirt, pin bore, top, and ring lands.

Inspect the piston pin for damage and wear.



Note : To ensure accuracy, the following measurements should be completed with the components at room temperature, 20°C [68°F].

Measure the piston skirt diameter 12 mm [0.5 in] from the bottom of the piston.



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B3.9 and B5.9 Piston Skirt Diameter

mm		in
101.823	MIN	4.0088
101.887	MAX	4.0107

B4.5 Piston Skirt Diameter

mm		in
101.833	MIN	4.0092
101.906	MAX	4.0120

Measure the piston ring clearance. Use a new piston ring to measure the clearance in the ring groove.

B4.5 Front Gear Train Piston Ring Clearance

	mm		in
Intermediate	0.040	MIN	0.0016
	0.110	MAX	0.0043
Oil Control	0.040	MIN	0.0016
	0.085	MAX	0.0033



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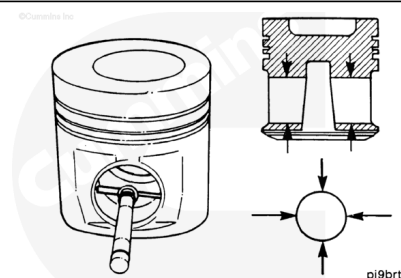
Note : The top piston ring clearance is not measured due to the type of piston ring used. The clearance can not be measured accurately with a typical feeler gauge.

Measure the pin bore.

Note : Measure the pin bore in the center. The bore profile is tapered on the edges and will give incorrect measurements.

B3.9 and B5.9 Piston Pin Bore Diameter

mm		in
40.006	MIN	1.5750
40.025	MAX	1.5758



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B4.5 Piston Pin Bore Diameter

mm		in
34.503	MIN	1.3584
34.522	MAX	1.3591

Piston Pin - Inspection

Inspect the piston pin for nicks, gouges, and excessive wear.

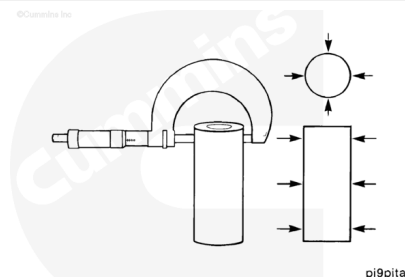
Measure the pin diameter.

B3.9 and B5.9 Pin Diameter

mm		in
39.990	MIN	1.5744
40.003	MAX	1.5749

B4.5 Pin Diameter

mm		in
34.483	MIN	1.3576
34.495	MAX	1.3581



Finishing Steps

- Install the piston and connecting rod assembly. Refer to Procedure 001-054, Piston and Connecting Rod Assembly (/qs3/pubsys2/xml/en/procedures/40/40-001-054.html).

